

Unit 10, Lesson 3: Exploring Absolute Value Equations



Benchmark

MA.912.AR.4.1 Given a mathematical or real-world context, write and solve one-variable absolute value equations.



Learning Targets

- ☐ I can describe solutions of an absolute value equation from a graph.
- ☐ I can solve absolute value equations.



10.3.1 Warm-Up: Critical Thinking

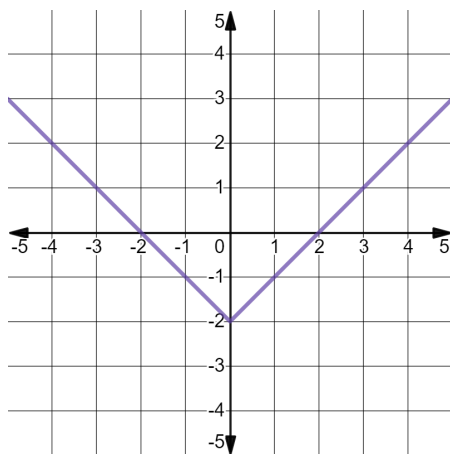
1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{} + \boxed{} = \boxed{} - \boxed{}$$



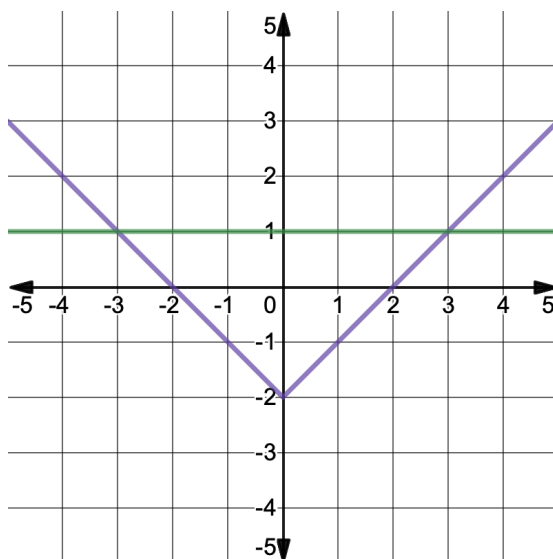
10.3.2 Exploration: Graphs of Absolute Value Functions

1. The graph of the absolute value equation $y = |x| - 2$ is shown.



The graph of $y = |x| - 2$ can be used to find the solution to $1 = |x| - 2$.

- a. What is the equation of the green line?



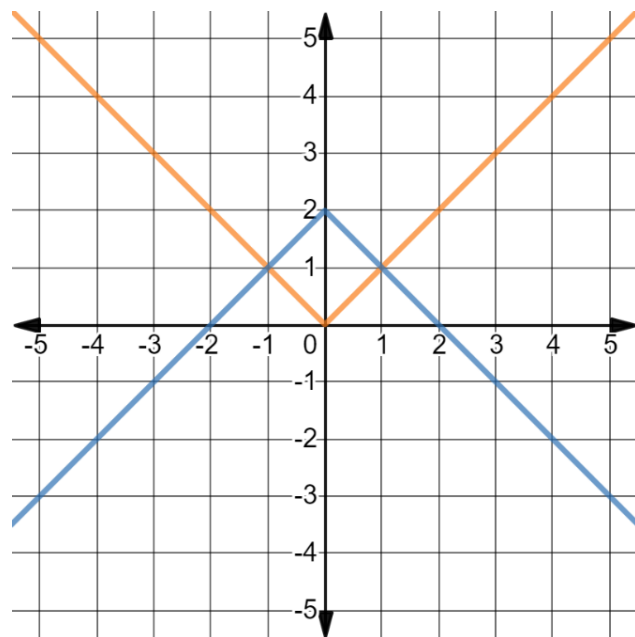
- b. What are the points of intersection for the two equations?

- c. Substitute the x -coordinates of the points of intersection into the equation $1 = |x| - 2$.

$1 = x - 2$	$1 = x - 2$
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- d. Discuss with your partner what you noticed when the x -coordinates were substituted into the equation. Write a summary of your discussion.

2. The graphs of $y = |x|$ and $y = -|x| + 2$ are shown on the coordinate grid.



- a. Decide who will be Partner A and who will be Partner B, then complete the table.
- b. What was true when there was only one value of x for the given y -value?
- c. What was true when there were two values of x for the given y -value?
- d. What was true when there were no values of x for the given y -value?
- e. Discuss with your partner the circumstances under which an absolute value function would have each type of solution.

Partner A		Partner B	
What is/are the value(s) of y when $x = 2$ on the blue graph?		What is/are the value(s) of y when $x = 0$ on the orange graph?	
What is/are the value(s) of x when $y = 1$ on the blue graph?		What is/are the value(s) of x when $y = 0$ on the orange graph?	
What is/are the value(s) of x when $y = 3$ on the blue graph?		What is/are the value(s) of x when $y = -3$ on the orange graph?	

3. Complete each sentence.

No Solution
There will be no solution when the y -value ...
One Solution
There will be one solution when the y -value ...
Two Solutions
There will be two solutions when the y -value ...

4. Sketch a graph for each solution type.

No Solution	One Solution	Two Solutions
		

5. Four absolute value equations are shown.

a. Work with your partner to determine the x -values that make the equations true.

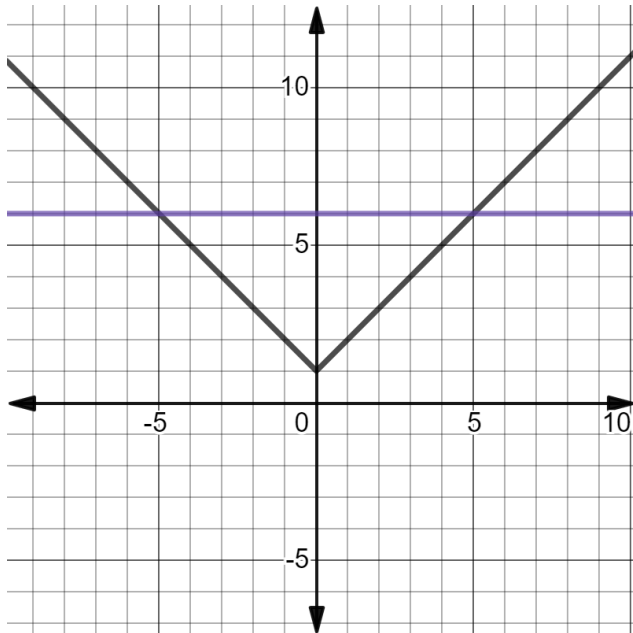
$x = 2 $	$x = -2 $
$ x = 3$	$ x = -3$

b. What did you notice about the four equations?



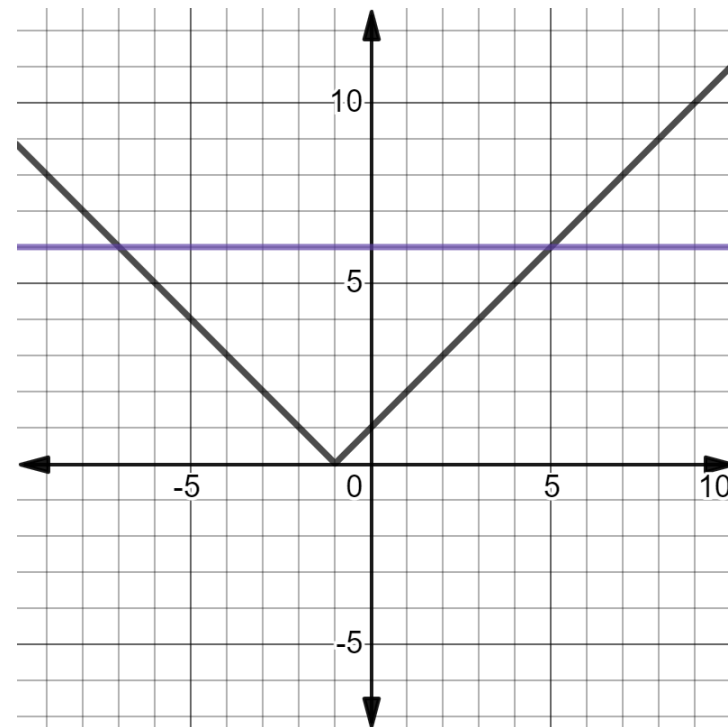
10.3.3 Guided Instruction: Solving an Absolute Value Equation

1. The graphs of $y = |x| + 1$ and $y = 6$ are shown on the coordinate grid.



- a. Use the graph to determine the solution to $|x| + 1 = 6$.
- b. Solve $|x| + 1 = 6$ algebraically.

2. The graphs of $y = |x + 1|$ and $y = 6$ are shown on the coordinate grid.



- a. Use the graph to determine the solution to $|x + 1| = 6$.
- b. Solve $|x + 1| = 6$ algebraically.



10.3.4 Guided Practice: Solving Absolute Value Functions

Solve each absolute value equation algebraically.

1. $|w| - 42 = -37$

2. $|x - 12| = 23$

3. $16 = |y| + 5$

4. $|9z| = 15$



10.3.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to them explaining how to solve absolute value equations.